

Statistics Unit 2: Effect Size and Error

Lecture Notes

What if Moniz conducted a clinical trial?

Leucotomy	BPRS score
Case 1, B.V.J.	27
Case 2, J.R.G.	36
Case 3, J.J.S.	54

$$\bar{x} = 39$$

Control	BPRS score
Case 4, C.W.K.	45
Case 5, K.S.H.	39
Case 6, K.K.T.	51

$$\bar{x} = 45$$

$$\text{Difference: } 39 - 45 = -6$$

How big is this effect?

Effect Size (descriptive)

A typical clinical trial for depression might find:

$$\text{Control: } \bar{x} = 26.5$$

$$\text{Cognitive therapy for depression: } \bar{x} = 17.0$$

$$\text{Difference: } 26.5 - 17.0 = \mathbf{9.5}$$

Outcome might be the Beck Depression Inventory

How big is this effect?

Effect size: “The effect is this big.”

NOT effect size: “The effect is not zero.”

Significance = “We had a big enough sample to determine the effect was not zero”

Research results in media and in text books

- There was an effect
- There was a correlation
- There was a difference
- The treatment is effective

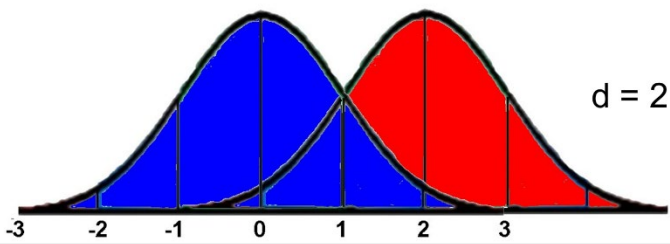
Hint: effect size cannot be answered “yes” or “no”

Informed consent to treatment: is it important?

Is it sufficient to say “Effect is not zero”

Do you want to know effect sizes for treatments you receive?

Can someone give informed consent without understanding the effect size?

$d = \frac{\bar{x}_1 - \bar{x}_2}{sd_{pooled}}$ 1 dichotomous variable 1 numerical variable  $d = 2$ Difference between two means in standard deviation units.	Cohen suggested cutoff scores $< .2$ = trivial $\geq .2$ = small $\geq .5$ = medium $\geq .8$ is large. $sd = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\frac{ss}{df}}$ $sd_{pooled} = \sqrt{\frac{ss_1 + ss_2}{df_1 + df_2}}$
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The d-statistic

A d-stat says how far apart two means are split measured with a standard deviation unit

Note regarding Cohen's *d*, Hedges' *g*, and Glass' *g*:

For this class, I will only use the term, "*d*-statistic." However, it would be good to know that the *d*-statistic is also called, "Cohen's *d*," and there are slight variations of this formula that involve different ways of calculating the pooled *sd*. The formula presented here matches the one provided by Cohen in his 1988 book "Statistical Power Analysis for the Behavioral Sciences."

This formula is also sometimes called a Hedges' *g*. According to Howell, Cohen initially expressed his formula only in terms of population parameters (*m* and *s*), whereas Hedges was the first to suggest using the pooled *sd* from a sample. Thus, Cohen's *d* is sometimes used to refer to a parameter, and then Hedges' *g* is used to refer to a statistic. Some sources attribute the above formula to Hedges, and then attribute to Cohen a different formula that uses an *unweighted* pooled *sd*. Sometimes, a *d*-statistic is called a Hedges' *g* if it has been corrected for bias. In small samples, *d* tends to overestimate the population effect, and Hedges developed a formula that corrects for this bias.

Glass' *g*' is similar to a Cohen's *d* (or Hedges' *g*). Instead of pooling standard deviations, Glass proposed using the standard deviation of the control group.

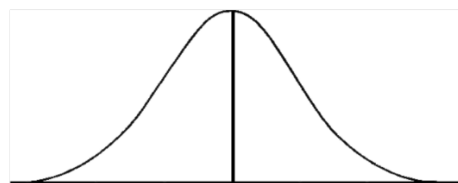
Meta analysis of **prolonged exposure** for posttraumatic stress disorder (PTSD)

13 clinical trials

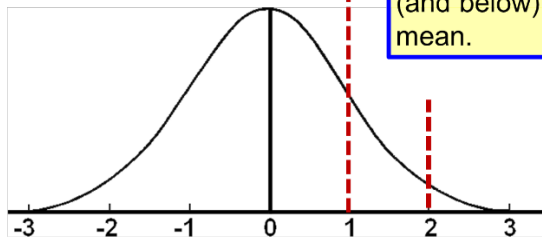
$d \approx 1.0$

Powers, M. B., Halpern, J. M., Ferenschak, M. P., Gillihan, S. J., & Foa, E. B. (2010). A meta-analytic review of prolonged exposure for posttraumatic stress disorder. *Clinical Psychology Review*, 30(6), 635-641.

How to draw the d-statistic effect size

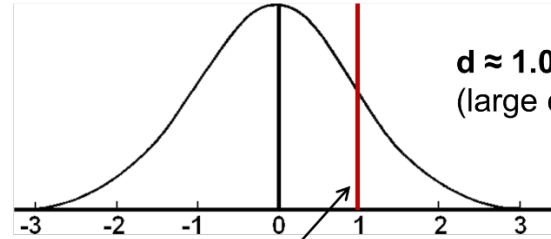


Draw a bell-shaped curve for the control group



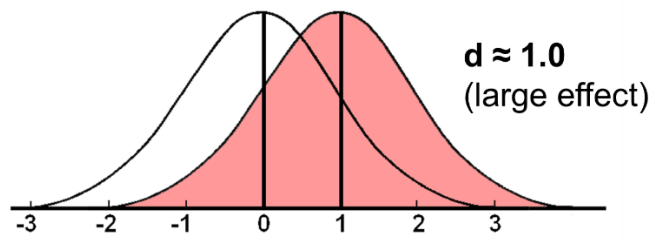
Note the location of 1 and 2 sd above (and below) the mean.

Mark off and number the standard deviation units



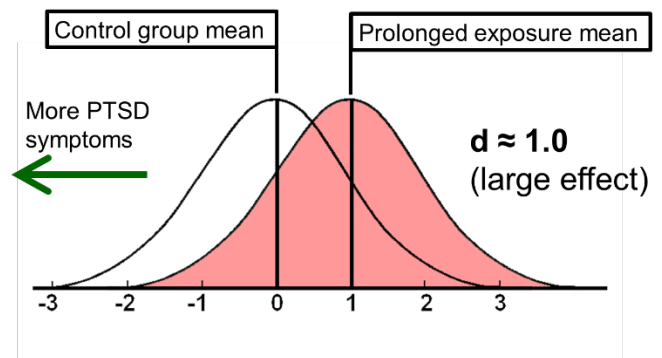
$d \approx 1.0$
(large effect)

Draw a line for the mean of the treatment group.



$d \approx 1.0$
(large effect)

Draw the curve for the treatment group.



$d \approx 1.0$
(large effect)

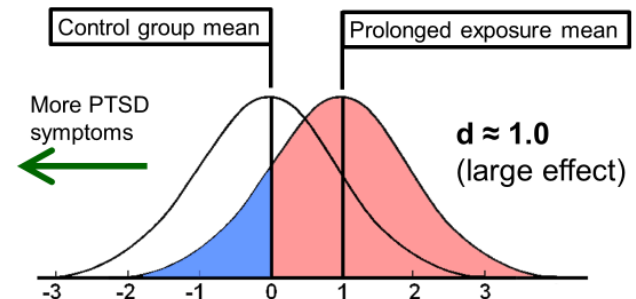
Label the means.
Label the direction of the outcome variable.

CL = Common Language effect size

CL gives the probability of superiority. It is the probability that, if you receive a treatment, you will be better off than someone without treatment.

Specifically, the probability that a randomly selected person from the treatment group will be better off than a randomly selected person from the control group.

Divide d by $\sqrt{2}$ and then look up the probability of obtaining a score lower than that value on the normal z -table.



$d \approx 1.0$
(large effect)

$$\frac{d}{\sqrt{2}} = \frac{1}{\sqrt{2}} = .707$$

CL = .76
round to .75

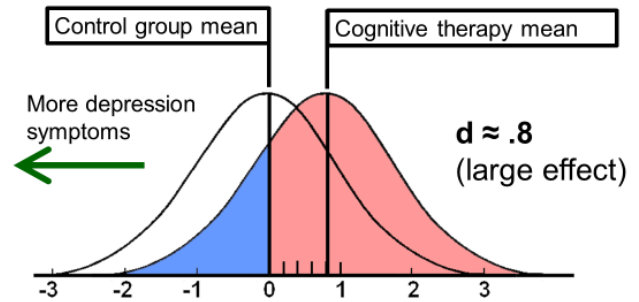
With **no** treatment your probability of superiority is **50%**,
with treatment it increases to **75%**,
which is an increase of **25%**.

Meta analysis of **Cognitive therapy** for depression

48 clinical trials

$d \approx .8$

Gloaguen, V., Cottraux, J., Cucherat, M., & Blackburn, I. (1998). A meta-analysis of the effects of cognitive therapy in depressed patients. *Journal of Affective Disorders*, 49(1), 59-72.



$$\frac{d}{\sqrt{2}} = \frac{.8}{\sqrt{2}} = .566$$

CL = .71
round to .70

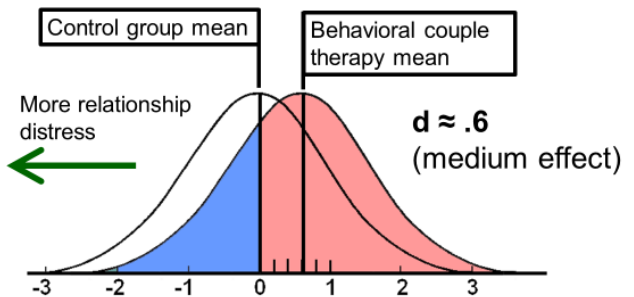
With **no** treatment your probability of superiority is **50%**,
with treatment it increases to **70%**,
which is an increase of **20%**.

Meta analysis of **Behavioral Couple Therapy** for relationship distress

30 clinical trials

$d \approx .6$

Shadish, W. R., & Baldwin, S. A. (2005). Effects of Behavioral Marital Therapy: A Meta-Analysis of Randomized Controlled Trials. *Journal Of Consulting And Clinical Psychology*, 73(1), 6-14.



$$\frac{d}{\sqrt{2}} = \frac{.6}{\sqrt{2}} = .424$$

CL = .66
round to .65

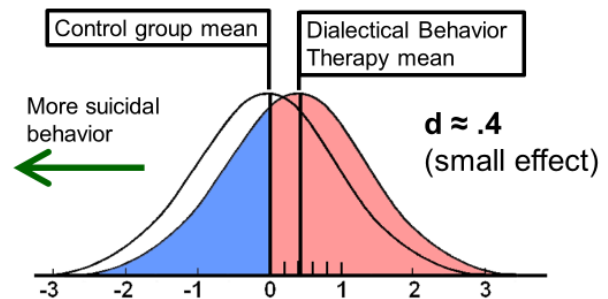
With **no** treatment your probability of superiority is **50%**,
with treatment it increases to **65%**,
which is an increase of **15%**.

Meta analysis of **Dialectical Behavior Therapy** for borderline personality disorder

8 clinical trials

$d \approx .4$

Kliem, S., Kröger, C., & Kosfelder, J. (2010). Dialectical behavior therapy for borderline personality disorder: A meta-analysis using mixed-effects modeling. *Journal Of Consulting And Clinical Psychology*, 78(6), 936-951.



$$\frac{d}{\sqrt{2}} = \frac{.4}{\sqrt{2}} = .283$$

CL = .61
round to .60

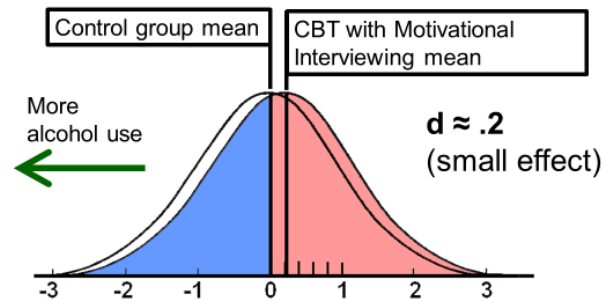
With **no** treatment your probability of superiority is **50%**,
with treatment it increases to **60%**,
which is an increase of **10%**.

Meta analysis of **Cognitive Behavior Therapy with Motivational Interviewing** for alcohol use disorder

9 clinical trials

$d \approx .2$

Riper, H., Andersson, G., Hunter, S. B., de Wit, J., Berking, M., & Cuijpers, P. (2014). Treatment of comorbid alcohol use disorders and depression with cognitive-behavioural therapy and motivational interviewing: A meta-analysis. *Addiction*, 109(3), 394-406.



$$\frac{d}{\sqrt{2}} = \frac{.2}{\sqrt{2}} = .141$$

CL = .56
round to .55

With **no** treatment your probability of superiority is **50%**,
with treatment it increases to **55%**,
which is an increase of **5%**.

Correlation

Strength of relationship

2 numerical variables

Range:

-1 perfect negative relationship

0 = no relationship

1 = perfect positive relationship

Similarity between variance and covariance

Variance

$$\frac{\sum (x - \bar{x})^2}{n - 1}$$

Covariance

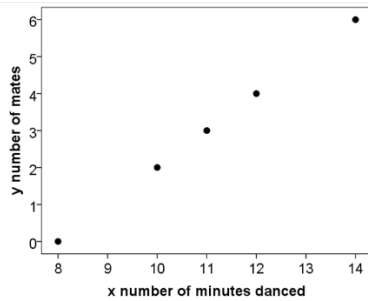
$$\frac{\sum (x - \bar{x}) (y - \bar{y})}{n - 1}$$

Covariance

- Will be as **large as possible** when. . .
- Will be as **negative as possible** when. . .
- Will **equal zero** when. . .
- Cannot be larger than. . .
- Will equal the averaged variance when. . .

Bird	x minutes	y mates
Polly	8	0
Flaps	10	2
Tweety	11	3
Big	12	4
Fried	14	6

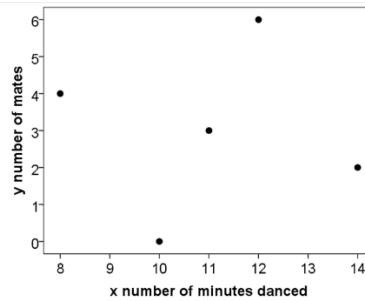
Bird	x minutes	y mates
Polly	8	4
Flaps	10	0
Tweety	11	3
Big	12	6
Fried	14	2



$$sd_x^2 = 5$$

$$sd_y^2 = 5$$

$$cov_{xy} = 5$$



$$sd_x^2 = 5$$

$$sd_y^2 = 5$$

$$cov_{xy} = 0$$

Variance and Covariance

Take your x and center it and times it by itself now.
 Divide that sum of squares you get by the sample minus one.
 Now you've got a variance, that's what you have done.
 Take your x and center it and times it by a partner.
 Divide that sum of cross products by the sample minus one.
 Now you've got covariance; x, y having fun.

How big is the covariance? How big could the covariance be?

Correlation = covariance / averaged variance

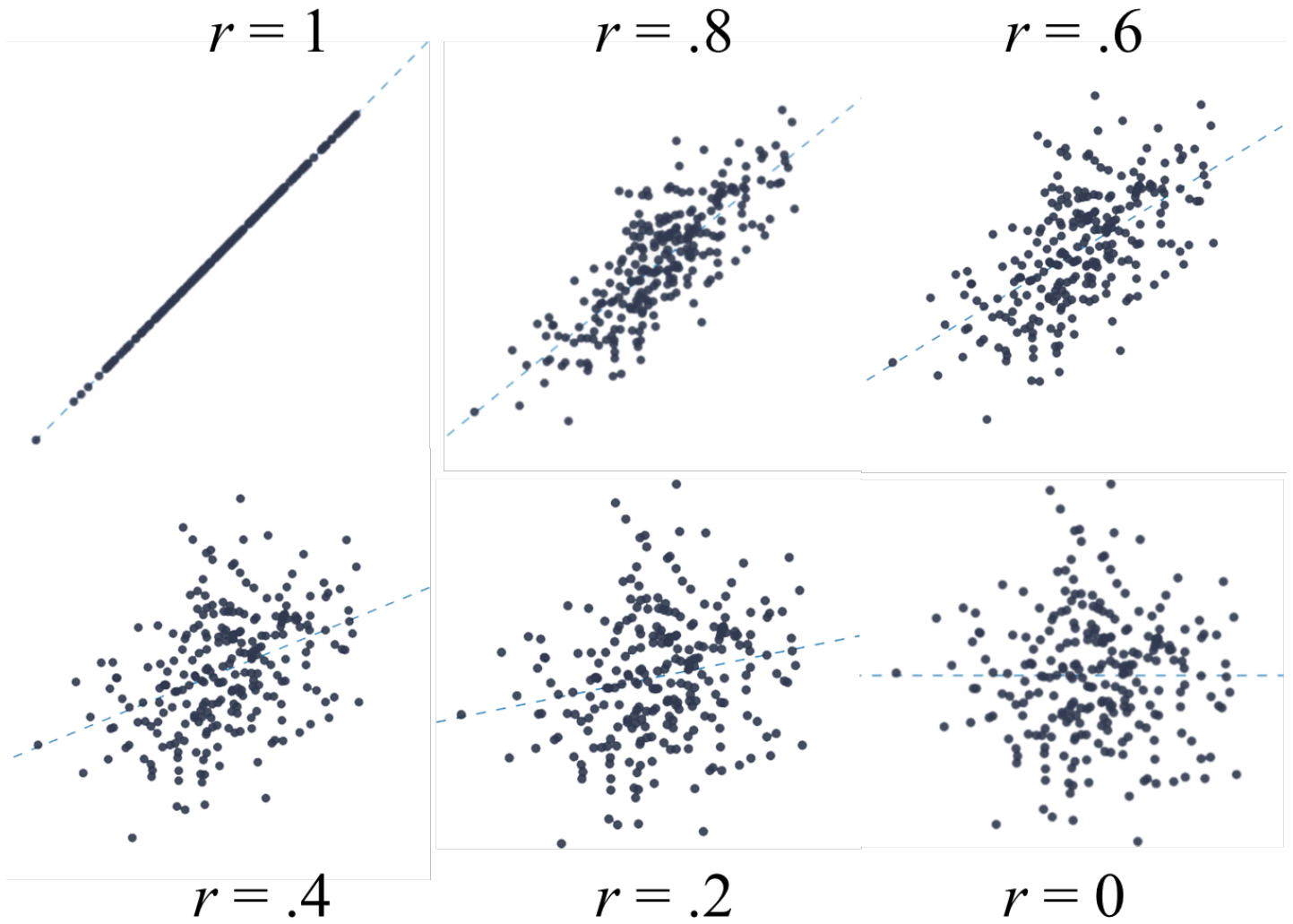
$$r_{xy} = \frac{Cov_{xy}}{sd_x sd_y}$$

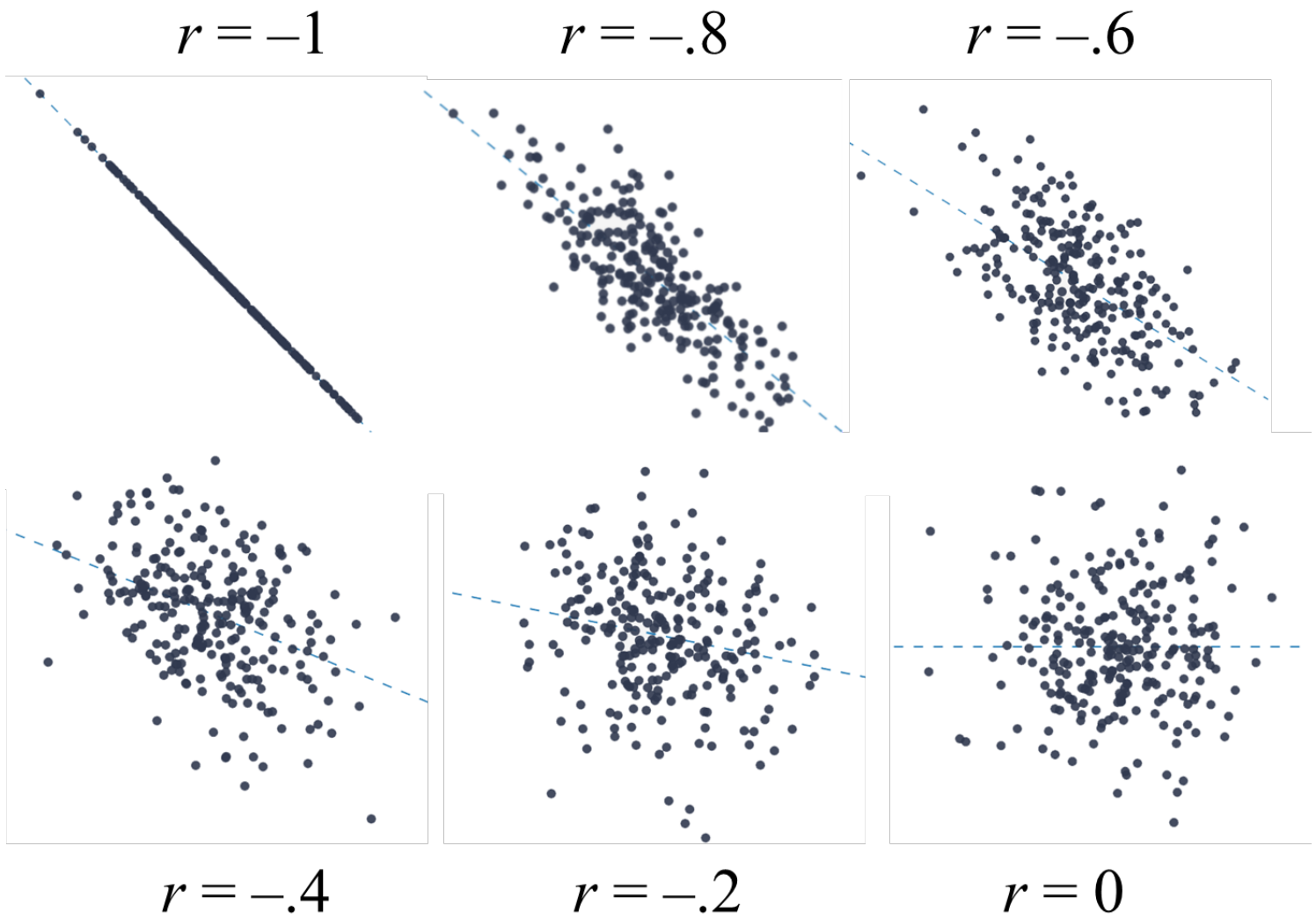
- Note: $sd_x sd_y$ is actually the geometric mean of the two variances.
- Thus, correlation is the covariance divided by a measure of averaged variance.

If both x and y are z scores ($sd = 1$), then. . .

$$r_{xy} = Cov_{xy}$$

Scatter plot examples:





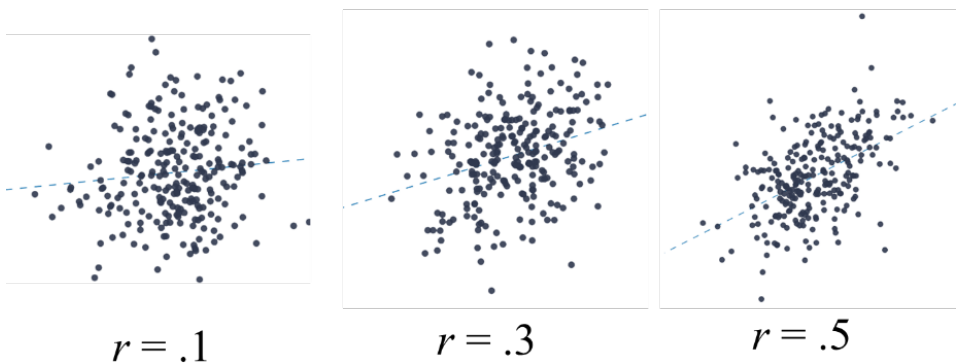
Cohen suggested cutoff scores

$< .1$ = trivial

$\geq .1$ = small

$\geq .3$ = medium

$\geq .5$ is large.

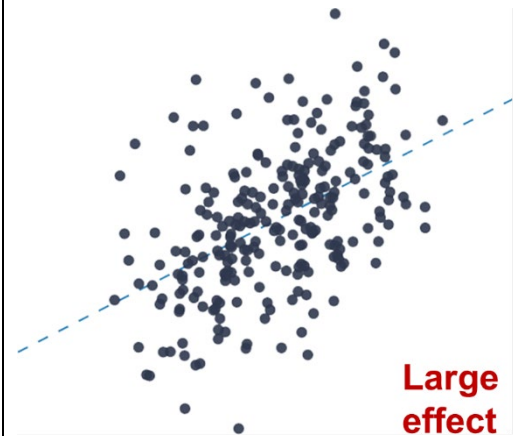


Meta analysis of association between **self compassion** and **psychopathology**

14 studies

$$r \approx .5$$

MacBeth, A., & Gumley, A. (2012). Exploring compassion: A meta-analysis of the association between self-compassion and psychopathology. *Clinical Psychology Review*, 32(6), 545-552.

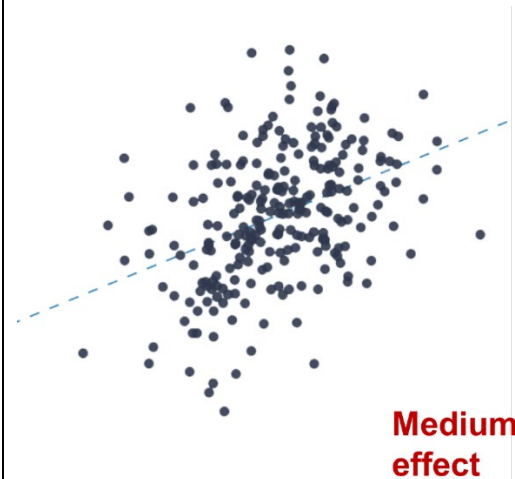


Meta analysis of association between **marriage quality** and **well-being**

137 studies

$$r \approx .4$$

Proulx, C. M., Helms, H. M., & Buehler, C. (2007). Marital quality and personal well-being: A meta-analysis. *Journal Of Marriage And Family*, 69(3), 576-593.

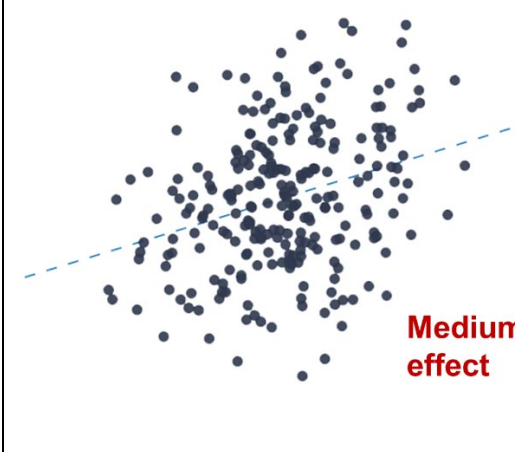


Meta analysis of association between **therapeutic alliance** and **psychotherapy outcome**

201 studies

$$r \approx .3$$

Horvath, A. O., Del Re, A. C., Flückiger, C., & Symonds, D. (2011). Alliance in individual psychotherapy. *Psychotherapy*, 48(1), 9-16.



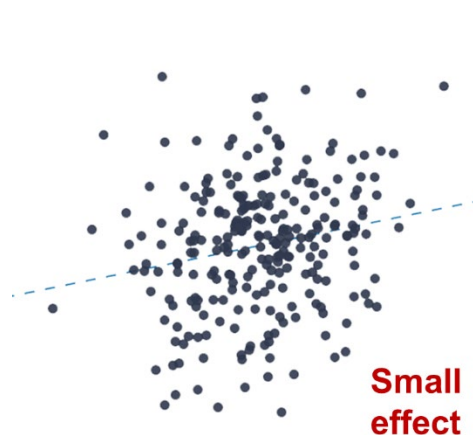
Meta analysis of association between **parenting** and **child delinquency**

161 studies

$$r \approx .2$$

Parenting examples: supportive parenting and monitoring

Hoeve, M., Dubas, J. S., Eichelsheim, V. I., van der Laan, P. H., Smeenk, W., & Gerris, J. M. (2009). The relationship between parenting and delinquency: A meta-analysis. *Journal Of Abnormal Child Psychology*, 37(6), 749-775.



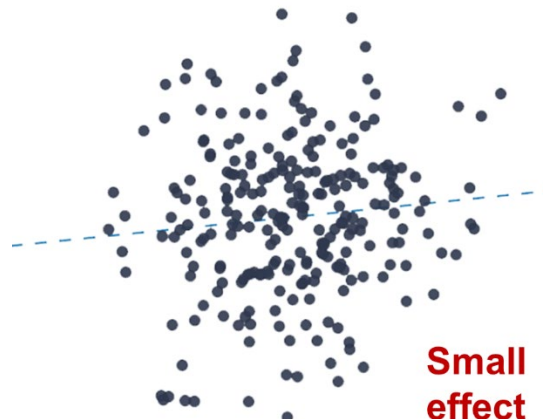
Meta analysis of association between **clinician confidence** and **accuracy**

36 studies

$$r \approx .1$$

Examples: confidence in diagnosis as verified by autopsy, future institutionalization as verified by legal or psychiatric records, client improvement as measured by objective tests.

Miller, D. J., Spengler, E. S., & Spengler, P. M. (2015). A meta-analysis of confidence and judgment accuracy in clinical decision making. *Journal Of Counseling Psychology*, 62(4), 553-567.



Types of distributions



	Measurement error	Standard deviation	Estimation error	Sampling error
Observations of what from what?	assessments from a case	cases from a sample	cases from a sample	samples from a population
Deviations around what?	true score	sample mean	predicted value	population parameter

case = person = individual

Standard deviation



The average person deviates from the **mean** this much
Linear equations (note the subscripts)

$$\hat{x}_{is} = \bar{x}_s$$

$$x_{is} = \bar{x}_s + e_{is}$$

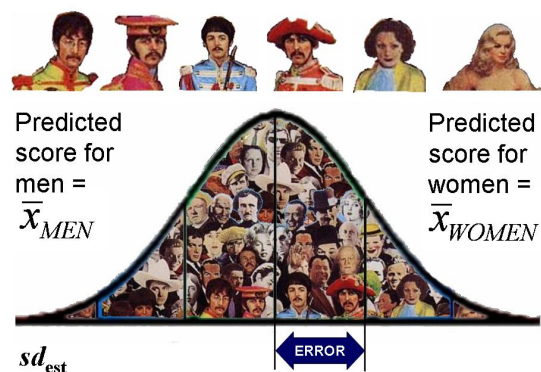
individual i from
sample s

average deviation of $|e_{is}|$

Standardization
$$z = \frac{x - \bar{x}}{sd}$$

The distribution is sometimes normal

Standard error of estimate (estimation error)



The average person deviates from **prediction** this much

Linear equations (note the subscripts)

$$\hat{y}_{is} = (\text{prediction}_s \text{ based on } x_{is})$$

$$y_{is} = (\text{prediction}_s \text{ based on } x_{is}) + e_{is}$$

y = rating how much you like Blue Footed Boobys (1 to 10)

x = gender

Predictions for sample “s” = if male, predict 5, if female, predict 7

Scores from sample “s”

Person i	y_{is}	x_{is}	$\hat{y}_{is}$	e_{is}
Lou Segusi	3	m	5	-2
Bud Uronner	9	m	5	4
Stu Earley	3	m	5	-2
Freida Gogh	6	f	7	-1
Sasha Noyes	8	f	7	1
Ophelia Paine	7	f	7	0

There is an inverse relationship between effect size (accuracy in prediction) and estimation error.

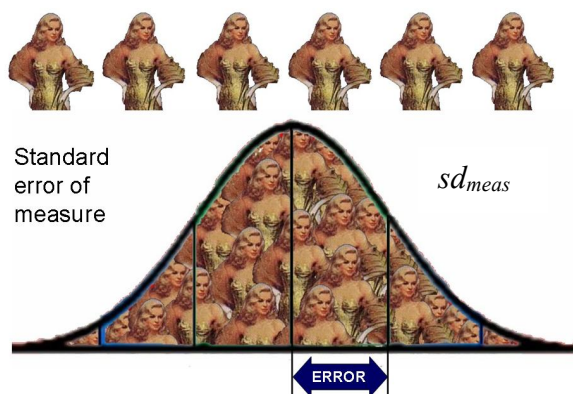
prediction “s” based on x_{is}

sd_{est} is the sd of e_{is}

The standard error of estimate

The standard error of estimate, tell me what it’s for.
Got people in a sample and you guess their scores.
It’s your typical mistake, no less no more.
It’s your typical mistake, no less no more.

Standard error of measure (measurement error)



Her average assessment deviates from **true** this much

Definition of a “true” score: A person’s hypothetical average score if he or she were repeatedly assessed using an infinite pool of assessment options.

Linear equations (note the subscripts)

$$\begin{aligned}\hat{x}_{ai} &= \text{true}_i \\ x_{ai} &= \text{true}_i + e_{ai}\end{aligned}$$

assessment occasion a
of individual i

average deviation of $|e_{ai}|$

Total variance – measurement error variance = true variance

Reliability

- true variance / total variance
- Percent of variance that is true
- Estimated by: $r_{\text{test1}, \text{test2}}$

Types of reliability

- Alternate forms
- Test retest
- Split half
- Cronbach's α

Arbitrary cutoffs:

- .80 = good
- .70 = adequate
- < .70 = poor



$$sd_{Meas}^2 = 1 - \text{reliability}$$

$$sd_{Meas} = \sqrt{1 - \text{reliability}}$$

$$sd_{Meas} = sd \sqrt{1 - \text{reliability}}$$

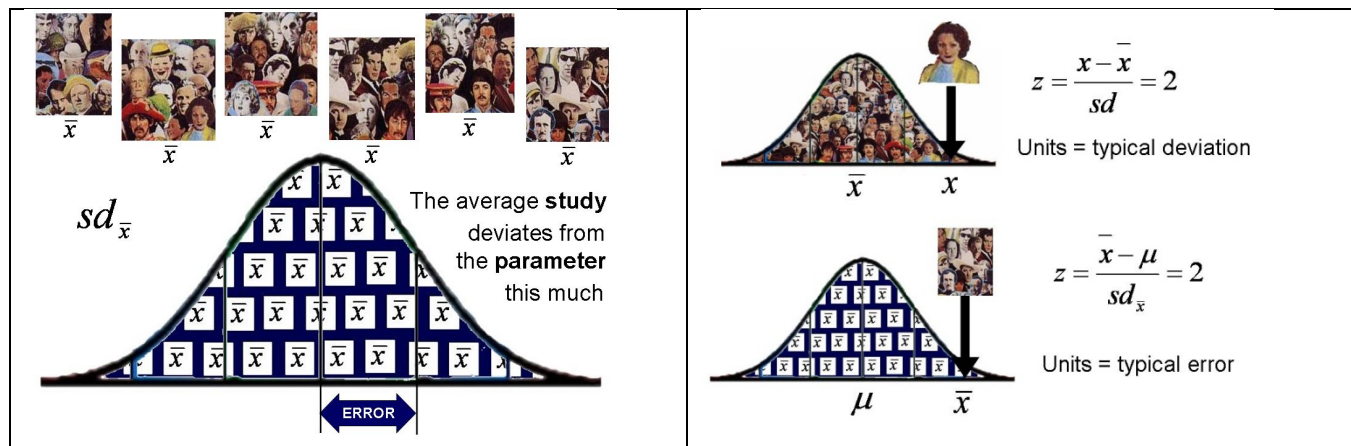
The standard error of measure

If you test me many times

my scores will change and stray from true (repeat)

The standard error of measure says just how far they do

Standard error of a statistic (sampling error)



Linear equations (note the subscripts)

$$\hat{x}_{sp} = \mu_p$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

sample s from
population p

$$(\widehat{\bar{y}} - \bar{y})_{sp} = (\mu - \mu)_p$$

$$(\bar{y} - \bar{y})_{sp} = (\mu - \mu)_p + e_{sp}$$

$$\hat{r}_{sp} = \rho_p \quad r_{sp} = \rho_p + e_{sp}$$

oddball

$$\hat{\beta}_{sp} = \beta_p \quad \beta_{sp} = \beta_p + e_{sp}$$

average deviation of $|e_{sp}|$

The standard error of a statistic will be normal when. . .

Influenced by n

Types of standard error:

mean, mean difference, correlation, beta

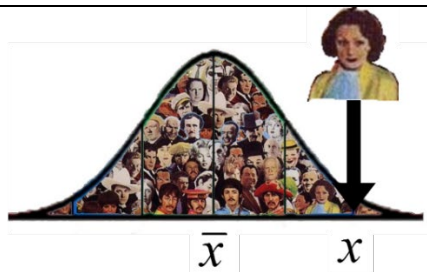
The standard error of a statistic

How much change do you typically find

If a study is repeated many times?

Each statistic we declare will have its own standard error.

Standard error of a statistic and significance testing

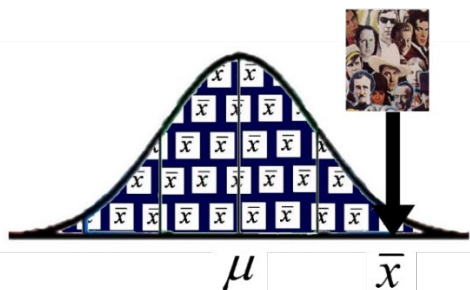


We can standardize
 x_i based on sd units

This person scored 2 standard deviations higher than the sample mean.

$$z = \frac{x - \bar{x}}{sd} = 2$$

Units = typical deviation



Can we standardize
 $\bar{x}_{sp}$ based on $sd_{\bar{x}}$ units?

Why would this be
useful?

The mean for this study is 2 standard errors higher than the population parameter.

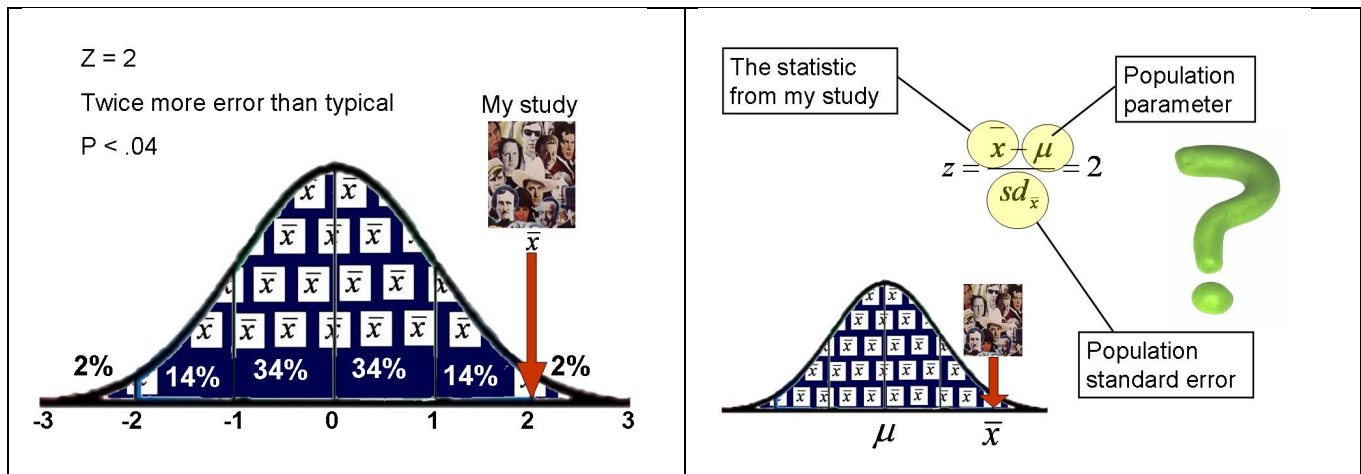
$$z = \frac{\bar{x} - \mu}{sd_{\bar{x}}} = 2$$

Units = typical error

Standardized sampling error

$$z = \frac{\text{statistic} - \text{parameter}}{sd_{\text{statistic}}}$$

$$z = \frac{\text{error in my study}}{\text{expected error}}$$



Where to get the population standard error?

T distribution:

$$t = \frac{\bar{x} - \mu}{sd_{\bar{x}}} = 2$$

Standard error estimated from sample

$$z = \frac{\bar{x} - \mu}{sd_{\bar{x}}} = 2$$

Population standard error

1. Standard error estimated from a sample
2. Degrees of freedom
3. Slightly flatter than z when n is small
4. Same as z when n is large

$$t = \frac{\text{statistic} - \text{parameter}}{sd_{\text{statistic}}}$$

Where to get the population parameter?

$$t = \frac{\bar{x} - \mu}{sd_{\bar{x}}} = 2$$

Make an assumption

Assume the parameter equals zero

Example: Assume I calculate the mean and standard error and that $t = 2$. This means. . .

If $\mu = 0$, then:

1. This study made twice the typical error.
2. The probability of making this much error is .04.

$$t = \frac{\text{statistic} - \text{parameter}}{sd_{\text{statistic}}}$$

Assume 0

$$t = \frac{\text{statistic} - 0}{sd_{\text{statistic}}}$$

$$t = \frac{\text{statistic}}{sd_{\text{statistic}}}$$

The t -value

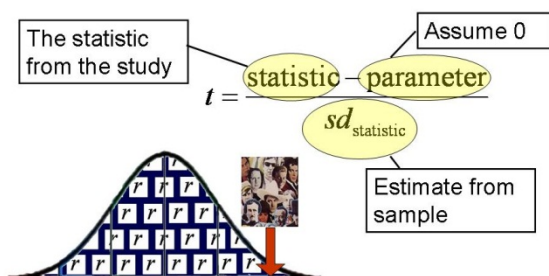
A t -value is always where you divide a stat by its standard error

Significance testing

Step 1: Assume the null hypothesis is true

Significance testing

Step 2: Collect data, calculate test statistic



Critique of hypothesis testing

- Power
- Confidence intervals
- Is H_0 ever really true?
- Effect size

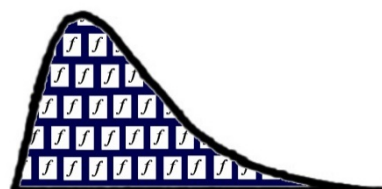
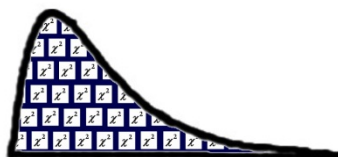
Step 3:

Determine probability (assuming H_0 is true).

Usually $\alpha = .05$

Distributions with a known shape when H_0 is true:

t χ^2 f



Significance

When the p is less than 0.05
I call it significant because I derive
the sample was big enough to make me a hero
and now I can claim the effect is not zero

Lecture Add-On Notes

Decision	True State of the World	
	H ₀ True	H ₀ False
Reject H ₀	Type I error $P = \alpha$	Correct decision $P = 1 - \beta = \text{power}$
Don't reject H ₀	Correct decision $P = 1 - \alpha$	Type II error $P = \beta$

Reliability Statistics

Cronbach's Alpha	N of Items
.93	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
sat_231	10.06	22.296	.797	.925
sat_232	9.88	21.406	.865	.912
sat_233	9.74	21.755	.860	.913
sat_234	9.64	22.652	.813	.922
sat_235	10.25	21.479	.800	.925

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
12.4	33.7	5.8	5

$$sd_{Meas} = 5.8\sqrt{1 - .93} = 1.53$$

Written summary

The satisfaction scale had good reliability (Cronbach's Alpha = .93). These results suggest that 7% of the variance was measurement error and that on average, a participant's observed score is likely to deviate from his or her true score by 1.53 units.

1. Name the scale
2. Describe as poor, adequate, or good
3. State alpha
4. Interpret percent of measurement error variance
5. Interpret standard error of measure.

A researcher analyzes data for a study and obtains the following:

$$r = .4 \text{ and } sd_r = .08$$

1. What is the t -value?
2. In the t -statistic formula, what is the assumed value for the population parameter?
3. Given the assumed value for the population parameter, what is the value that indicates the amount of sampling error that was made in this study?
4. If this study were repeated over and over, what is the typical amount of sampling error that would be made in each study?
5. Given the assumed value for the population parameter, the error in this study is equal to the typical error times what value?
6. One component of the t -formula is something that is calculated from the sample, and if we knew the true population parameter for this, there would be no reason for doing the t -test. In the above example, what is this value?
7. Another component of the t -formula is something that is calculated from the sample, and if we knew the true population parameter for this, the distribution would change from a t -distribution to a z -distribution. In the above example, what is this value?